

Jevi Waugh

☎ 0420 463 525 • ✉ schallum0712@gmail.com • 🌐 Jevi-Waugh
in jevi-waugh • ORCID: jevi.waugh@uq.edu.au Website: jevi-waugh.github.io

Education

The University of Queensland

Master of Data Science

St Lucia, Brisbane

2026 - Current

Relevant coursework: Data Analytics at Scale (SQL, PIG, PySpark)

The University of Queensland

Bachelor of Computer Science (Machine Learning Major)

St Lucia, Brisbane

2024 - 2025

Semester 2 GPA: 6.5, **Relevant coursework:** Statistical Learning, Pattern recognition and Analysis

Experience

The University of Queensland

Research Assistant

School of EECS

March 2026 - Current

- Deriving the generalised Kaleidoscope transform to use in frequency based autoencoders for segmentation.

The University of Queensland

Summer Research Scholar

School of EECS

Jan–March 2026

- Designed and implemented a **PsychoNet-based decoder** for medical image segmentation, deriving the **Fourier Shuffle algorithm** for frequency–spatial upsampling; contributing author on the research paper *Learning Frequency Domain Codes for Semantic Vision*, scheduled for publication on May 8.
- Implemented and evaluated the architecture in **PyTorch**, benchmarking against the **state-of-the-art (SOTA) U-Net model**, achieving improved validation performance and lower GPU memory usage while running experiments on GPU compute nodes using custom training scripts. Supervised by Shekhar (Shakes) Chandra and Wendi Ma

The University of Queensland

Casual Academic

School of EECS

July 2025 - Current

- Casual Academic for **COMP4702/7703 (Machine Learning)**, **COMP3400 (Functional and Logic Programming)**, **COMP2048 (Theory of Computing)**, and **CSSE1001 (Introduction to Software Engineering)**.
- Deliver classes, guide students through programming and concepts, and assist with assessment marking and feedback.

UQ Computing Society

Industry Officer

UQ Union

Nov 2025 - Current

- Lead the UQCS sponsorship subcommittee, securing partnerships through industry outreach, pitching, and negotiation.
- Increased sponsorship funding by **40% in 2026** while expanding the sponsor base by **30%**.
- Onboarded new industry partners: **Cybermetrix**, **Quantum Australia**, **Atlassian**, **Tanda**, **Citadel**, and **getcracked.io**.
- Developed the annual sponsorship prospectus and maintain relationships with industry partners supporting student initiatives.

Technical Skills and Frameworks

Programming Languages: Python, Haskell, C (ANSI C), Java and SQL **Tools:** Git (Github), Linux, Jira

Libraries and Frameworks: PyTorch, Sk-learn, Einops, OpenCV, Pandas, NumPy, Matplotlib

Projects

Vision Transformer Attention Analysis

July – December 2025

- Investigated internal attention dynamics of Vision Transformer (ViT) architectures using pretrained DeiT models.
- Built a pipeline to extract and analyze attention distributions across transformer layers.
- Conducted experiments modifying attention formulations to study behaviour in collaboration with Dr. Kasra Khosoussi.

Fine-tuning FLAN-T5 for Biomedical Lay Summarisation

- Fine-tuned **FLAN-T5** encoder–decoder models on the **BioLaySumm** dataset using mixed-precision training.
- Evaluated model outputs using **ROUGE-1**, **ROUGE-2**, **ROUGE-L**, and **ROUGE-Lsum** metrics.
- Achieved **>70% ROUGE performance** across evaluation metrics on held-out biomedical summarisation tasks.

TraceForest: AI-Generated Code Detection

- Developed an explainable Random Forest classifier achieving **94% accuracy** for detecting AI-generated code using stylometric AST features. Integrated **SHAP-based explainability** to identify feature contributions influencing model predictions.
- Built Python and Java adapters returning classification probabilities for automated pull-request review workflows.
- Extended CodeGPTSensor via encoder fine-tuning with stylometric features, achieving **>95% accuracy**.
- Released a developer tool as a **VS Code extension** with **1.5K+ downloads**. Marketplace link